NURSING INFORMATICS EDUCATION FOR DIGITAL HEALTH

Findings from a Scoping Review and Proposing Future Directions

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SCOPING REVIEW: WHAT IS CURRENTLY BEING TAUGHT?

WHY THIS REVIEW?

Digital systems—from EHRs and virtual care to AI, predictive analytics and patient-generated data—are reshaping nursing work. Competency frameworks identify what nurses should know, but there is less clarity about how those competencies are taught across education and practice.

METHODS

- Scoping review of nursing informatics competency education
- Undergraduate, graduate and professional nursing education
- Two independent reviewers; thematic analysis of teaching strategies mapped onto CASN (Canadian Association of Schools of Nursing) domains
- 6 databases: Scopus, Web of Science, CINAHL, ERIC, Embase, Medline

AIM

Map teaching strategies and use the findings to identify priorities for a more integrated, competency-based and systems-oriented model.

WHAT THE LITERATURE EMPHASIZES—AND WHAT IT MISSES

COMMONLY TAUGHT

- Documentation and EHR workflow navigation
- Medication administration and information retrieval
- Privacy, confidentiality and responsible data use
- Foundational patient-safety and digital practice skills
- Information literacy and evidence-informed practice

UNDER-REPRESENTED

- Artificial intelligence and predictive analytics
- Patient-generated and real-world data
- Algorithmic bias, digital equity and data governance
- Effects on nursing judgment, relationships and systems of care
- Nurses' roles in technology design and implementation

IMPLICATION

Nursing informatics education remains uneven and strongly EHR-centred.

The next step is to preserve foundational skills while deliberately expanding what nurses learn to question, evaluate, improve and lead in digitally enabled care.

FUTURE DIRECTIONS: FROM DIGITAL TOOL USE TO SYSTEMS LEADERSHIP

A proposed agenda derived from the review findings and knowledge-exchange discussions

THE GOAL — **Prepare nurses to use, question, improve and lead within digitally enabled care environments.**

PROPOSED EDUCATIONAL TRAJECTORY

1 FOUNDATIONS

Use digital systems safely

Document & retrieve information

Protect privacy & data

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2 CRITICAL APPLICATION

Interpret data & AI outputs

Recognize bias and limitations

Integrate evidence + context

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3 SYSTEMS LEADERSHIP

Co-design and improve technology

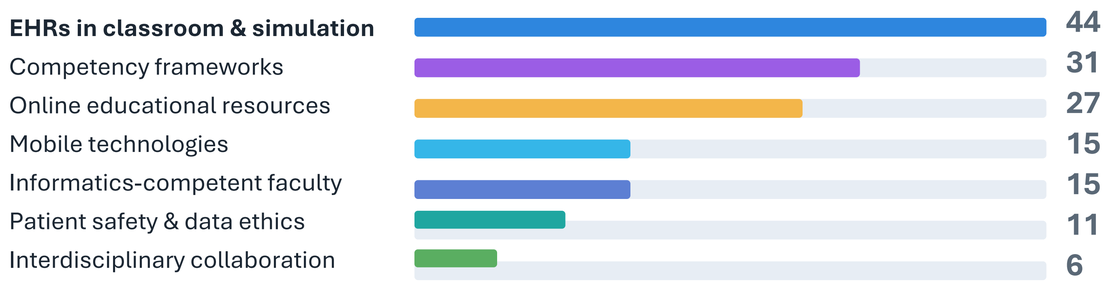
Shape workflows & governance

Lead implementation and change

REVIEW AT A GLANCE

120	7	6	3
PUBLICATIONS	TEACHING STRATEGIES	DATABASES	EDUCATION LEVELS

SEVEN TEACHING STRATEGIES IDENTIFIED



SIX PRIORITIES FOR PREPARING NURSES FOR TECHNOLOGY-ENABLED PRACTICE

01 Advance Informatics Education

- Scaffold across the cirriculum by introducing informatics early, then revisit it as clinical experience grows
- Introduction to clinical decision support tools; small group discussion of case studies that focus on breach in privacy;
- Contemporary issues of digital inequity and Indigenous populations; assess access barriers

02 Develop clinical judgement and nursing skills

- Use technology to augment—not replace—clinical reasoning (e.g. critique AI generated plan)
- Address automation bias, cognitive offloading, deskilling
- Require learners to verify and contextualize technology-generated recommendations

03 Build critical AI and data literacy

- Connect technology to assessment, prioritization, documentation and care coordination
- Examine alerts, multiple systems, downtime and workflow disruption
- Use clinical experiences to identify where technology helps—or hinders—care

04 Promote application to practice

- Ground the learning into the nursing workflow
- Identify clinical problems that could benefit from digital solutions and critique them
- Case studies and post conference debriefs about technology

05 Strengthen academic–practice partnerships

- Co-design learning with partners
- Critique usability, workflow fit, equity, safety and unintended consequences
- Create digital-health practicums and QI/implementation projects
- Feed practice needs back into curriculum updates

06 Prepare for emerging technologies

- Build transferable skills for evaluating rapidly evolving technologies
- Be prepared for new technologies and advancements like vibe coding, explore generative AI, AI agents, etc
- Pair experimentation with safety, evidence, ethics, equity and governance

A SYSTEMS-ORIENTED MODEL: ALIGN CURRICULUM, ORGANIZATIONS AND PRACTICE

LEARNER

- Progressive competencies
- Critical digital literacy
- Ethical reasoning

CURRICULUM

- Longitudinal scaffolding
- Authentic assessment
- Cross-course integration
- Interdisciplinary

ORGANIZATION

- Faculty capability
- Digital infrastructure
- Protected development time

PRACTICE + SYSTEM

- Academic–practice partnership
- Co-design + implementation
- Governance and quality improvement

Reference: Chu CH, Siganevich Y, Donato-Woodger S, et al. Teaching Informatics Competencies within Nursing Education: A Scoping Review of Teaching Strategies. Applied Clinical Informatics. 2025;16(4):911–929. doi:10.1055/a-2648-4914